All that was bad news not only for the Soviet missile program but also for Khrushchev personally. He had gambled on missiles, cutting all other branches of his military significantly. But he had no solid-fuel rocket engines, which meant that his rockets had to spend hours on launchpads to be filled with liquid fuel, making them vulnerable to American strikes. They could be used as first-strike weapons but were useless for retaliation. “Father looked around the room gloomily. The result he wanted had once again proved impossible to achieve,” remembered Sergei Khrushchev. “He asked those present to think about what could be done to reduce to a minimum the amount of time it would take to catch up with the Americans.” The Soviets were behind not just in quantity of missiles but also in their quality, which meant that the missile gap long known to Khrushchev had just become wider.

The Soviet leader demanded new ideas and projects. He was eager to fill the gap as soon as possible. On April 16, a few days after Khrushchev and his colleagues in the party Presidium authorized a major military aid package for Cuba, they approved a new program to build intercontinental ballistic missiles capable of delivering nuclear warheads (euphemistically called a “special charge” in the party and government resolution) to the United States. Khrushchev’s new favorite, the missile designer Vladimir Chelomei, was authorized to develop two new missiles—the Universal Rocket 500, capable of delivering a 50-megaton charge equivalent to the Tsar Bomba, and the Universal Rocket 200, capable of carrying a 7,000-pound payload a distance of up to 12,000 kilometers. Mikhail Yangel in Dnipropetrovsk was charged with building a new rocket, the R-36, with a range of 16,000 kilometers. It would become known in the West as the SS-18 Satan, the first Soviet MIRV, or multiple independently targetable reentry vehicle missile. It could take a warhead into orbit and keep it there, reducing the time required to execute a strike to a few minutes.[12]

But all that would come later, with Satan being deployed only in 1974. Meanwhile, Khrushchev needed a solution. He started with a shakeup at the top of the Soviet missile establishment. Marshal Moskalenko, who had failed to keep the missile designers in line, had to go. At the Pitsunda meeting Sergei Korolev and the designer of the rocket engines, Valentin Glushko, got into a nasty fight over the type of fuel they wanted to use for a